

DBMON-01: Database Monitoring Fundamentals

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Overview

This notebook introduces how Dynatrace monitors databases across your environment. You will learn how OneAgent automatically detects database calls through distributed tracing, how database spans capture query-level detail, and how the Dynatrace entity model represents database services. We also cover ActiveGate extensions for remote database monitoring and the full range of supported database technologies.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with Grail enabled
OneAgent	Deployed on application hosts making database calls
Permissions	<code>storage:spans:read</code> , <code>storage:entities:read</code> , <code>storage:metrics:read</code>
Data	At least 1 hour of application traffic generating database calls

1. How Database Monitoring Works

Dynatrace monitors databases through two complementary approaches:

Approach	How It Works	What It Captures
OneAgent Code-Level Instrumentation	Automatically instruments database client libraries (JDBC, ADO.NET, database drivers)	Query text, execution time, result counts, connection details
ActiveGate Extensions	Connects remotely to database management interfaces (SQL queries against system tables)	Internal DB metrics: tablespace usage, lock waits, buffer cache hit ratio

OneAgent captures every database call as a **span** in the distributed trace. These spans contain:

- **db.system** — The database technology (e.g., `postgresql` , `mysql` , `mongodb`)
- **db.statement** — The query or command executed (normalized to remove literals)
- **db.operation** — The operation type (`SELECT` , `INSERT` , `UPDATE` , `DELETE` , `find` , `aggregate`)
- **db.namespace** — The database or schema name
- **server.address** — The database server hostname or IP
- **server.port** — The database server port

Note: Dynatrace normalizes SQL statements by replacing literal values with `?` placeholders. This groups identical query patterns together regardless of parameter values.

2. Database Span Anatomy

Let's examine the structure of a database span by querying for recent database calls and inspecting the available fields.

```
// Inspect database span fields – sample 10 recent DB calls
fetch spans, from:-1h
| filter isNotNull(db.system)
| fields timestamp, db.system, db.operation, db.namespace,
      db.statement, server.address, server.port,
      duration, span.kind, dt.entity.service
| sort timestamp desc
| limit 10
```

The query above returns the key attributes of each database span:

Field	Description	Example
<code>db.system</code>	Database technology identifier	<code>postgresql</code> , <code>mysql</code> , <code>mongodb</code>
<code>db.operation</code>	SQL command or DB operation	<code>SELECT</code> , <code>INSERT</code> , <code>find</code>
<code>db.namespace</code>	Database or schema name	<code>orders_db</code> , <code>inventory</code>
<code>db.statement</code>	Normalized query text	<code>SELECT * FROM orders WHERE id = ?</code>
<code>server.address</code>	Database host	<code>db-prod-01.internal</code>
<code>duration</code>	Execution time in nanoseconds	<code>1500000</code> (1.5ms)
<code>span.kind</code>	Always <code>CLIENT</code> for outgoing DB calls	<code>CLIENT</code>

3. Discovering Database Services

Dynatrace automatically creates **database service entities** when it detects database calls. These entities represent the logical database endpoint, not the host. Let's discover what database services exist in your environment.

```
// Discover all database service entities in the environment
fetch dt.entity.service
| filter serviceType == "DATABASE_SERVICE"
| fields entity.name, databaseHostNames, databaseVendor, softwareTechnologies
| sort entity.name asc
| limit 50

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | filter serviceType == "DATABASE_SERVICE"
// | fields name, databaseHostNames, databaseVendor, softwareTechnologies
// | sort name asc
// | limit 50
```

You can also discover databases through the spans themselves, which is useful when entity detection hasn't yet completed or when you want to see databases called from specific services.

```
// Discover database technologies from span data
fetch spans, from:-1h
| filter isNotNull(db.system)
| summarize call_count = count(),
             avg_duration_ms = avg(duration) / 1000000.0,
             distinct_statements = countDistinct(db.statement),
             by:{db.system, db.namespace, server.address}
| sort call_count desc
| limit 20
```

4. Database Span Exploration

Understanding the distribution of database calls helps identify which databases are most heavily used and where optimization efforts should focus.

```
// Database call volume by technology over the last hour
fetch spans, from:-1h
| filter isNotNull(db.system)
| summarize total_calls = count(),
             avg_duration_ms = avg(duration) / 1000000.0,
             p95_duration_ms = percentile(duration, 95) / 1000000.0,
             max_duration_ms = max(duration) / 1000000.0,
             by:{db.system}
| sort total_calls desc
```

```
// Database call volume over time – identify traffic patterns
fetch spans, from:-6h
| filter isNotNull(db.system)
| makeTimeseries call_count = count(), by:{db.system}, interval:5m
```

```
// Distribution of database operations (SELECT vs INSERT vs UPDATE vs DELETE)
fetch spans, from:-1h
| filter isNotNull(db.system) and isNotNull(db.operation)
| summarize op_count = count(), by:{db.system, db.operation}
| sort db.system asc, op_count desc
```

5. Database Types and Technologies

Dynatrace supports a broad range of database technologies through OneAgent auto-instrumentation:

Category	Technologies	db.system Values
Relational (SQL)	PostgreSQL, MySQL/MariaDB, Microsoft SQL Server, Oracle, IBM Db2	postgresql , mysql , mssql , oracle , db2
NoSQL Document	MongoDB, Couchbase, Amazon DynamoDB, Azure Cosmos DB	mongodb , couchbase , dynamodb , cosmosdb
NoSQL Key-Value	Redis, Memcached, Amazon ElastiCache	redis , memcached
NoSQL Column	Apache Cassandra, Apache HBase, ScyllaDB	cassandra , hbase
Search Engines	Elasticsearch, OpenSearch, Apache Solr	elasticsearch , opensearch , solr

Category	Technologies	db.system Values
Message Brokers	Apache Kafka, RabbitMQ, Amazon SQS	kafka , rabbitmq
Graph	Neo4j, Amazon Neptune	neo4j , neptune

Important: The `db.system` field follows the OpenTelemetry semantic conventions. The exact values may vary depending on the database driver and instrumentation version.

```
// Discover which database technologies are active in your environment
fetch spans, from:-24h
| filter isNotNull(db.system)
| summarize call_count = count(),
             unique_databases = countDistinct(db.namespace),
             unique_servers = countDistinct(server.address),
             by:{db.system}
| sort call_count desc
```

6. ActiveGate Extensions for Remote Monitoring

While OneAgent captures database calls from the application side (client spans), ActiveGate extensions monitor the database server itself. This provides internal metrics that are invisible from the application perspective.

Extension Architecture

Component	Role
ActiveGate	Hosts the extension runtime; must be host-based (not K8s-based) for Extensions 2.0
Extension 2.0	Python-based plugin that queries database system tables on a schedule
Monitoring Configuration	Defines connection string, credentials, and polling interval

Available Extensions

Extension	Metrics Captured
PostgreSQL	Connections, transactions/sec, tuple operations, table/index sizes, lock waits, replication lag
MySQL	Connections, queries/sec, buffer pool hit ratio, InnoDB row operations, slow queries
MS SQL Server	Batch requests/sec, page life expectancy, buffer cache hit ratio, deadlocks, wait stats
Oracle	Sessions, tablespace usage, SGA/PGA memory, redo log waits, library cache hit ratio
MongoDB	Connections, operations/sec, document metrics, replica set health, storage engine stats

Note: Extensions 2.0 require a **host-based ActiveGate**, not a Kubernetes-based deployment. See the Dynatrace documentation for extension installation procedures.

7. Baseline Database Metrics

Establishing a performance baseline is essential for detecting anomalies. The following queries help you understand your typical database performance characteristics.

```
// Database response time baseline – hourly P50, P95, P99 over the last 24 hours
fetch spans, from:-24h
| filter isNotNull(db.system)
| makeTimeseries p50_ms = percentile(duration, 50) / 1000000.0,
                  p95_ms = percentile(duration, 95) / 1000000.0,
                  p99_ms = percentile(duration, 99) / 1000000.0,
                  interval:1h
```

```
// Error rate baseline – database call failures over time
fetch spans, from:-24h
| filter isNotNull(db.system)
| makeTimeseries total = count(),
                  errors = countIf(otel.status_code == "ERROR"),
                  interval:1h
| fieldsAdd error_rate_pct = arrayAvg(errors) / arrayAvg(total) * 100
```

8. Summary and Next Steps

In this notebook you learned:

- How Dynatrace captures database activity through OneAgent span instrumentation
- The key span attributes (`db.system` , `db.statement` , `db.operation` , `db.namespace`) that describe each database call
- How to discover database service entities and active database technologies
- The role of ActiveGate extensions for server-side database metrics
- How to establish a performance baseline using span data

Next Steps

- **DBMON-02: SQL Database Monitoring** — Deep dive into relational database analysis with PostgreSQL, MySQL, MS SQL, and Oracle
- **DBMON-03: NoSQL Database Monitoring** — MongoDB, Cassandra, DynamoDB, and Cosmos DB analysis

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